



MAT1320-Linear Algebra Lecture Notes

Linear Dependence and Independence of Vectors and Spanning Sets

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Linear Dependence and Independence of Vectors

Linear Dependence and Independence of Vectors

Definition

Let V be a real vector space, $x_1, x_2, \dots, x_n \in \mathbb{R}$ and

$$\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \in V.$$

Linear Dependence and Independence of Vectors

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$\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \in V$. We say vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are **linearly dependent** if there exists scalars $x_1, x_2, \dots, x_n \in \mathbb{R}$, not all of them zero, such that $x_1 \vec{v}_1 + x_2 \vec{v}_2 + \dots + x_n \vec{v}_n = \vec{0}$.

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Note: If the only solution of the homogeneous system

$x_1 \vec{v}_1 + x_2 \vec{v}_2 + \dots + x_n \vec{v}_n = \vec{0}$ is zero solution, then we say vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly independent.

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Note: If the only solution of the homogeneous system $x_1 \vec{v}_1 + x_2 \vec{v}_2 + \dots + x_n \vec{v}_n = \vec{0}$ is zero solution, then we say vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly independent. If the homogeneous system $x_1 \vec{v}_1 + x_2 \vec{v}_2 + \dots + x_n \vec{v}_n = \vec{0}$ has a nonzero solution, then we say vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly dependant.

Linear Dependence and Independence of Vectors

Example

The vectors

$\{\vec{e}_1 = (1, 0, 0), \vec{e}_2 = (0, 1, 0), \vec{e}_3 = (0, 0, 1)\} \subset \mathbb{R}^3$ are linearly independent.

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$$\Rightarrow x_1 \vec{e}_1 + x_2 \vec{e}_2 + x_3 \vec{e}_3 = (0, 0, 0)$$

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$$\Rightarrow (x_1, x_2, x_3) = (0, 0, 0)$$

$$\Rightarrow x_1 = 0, x_2 = 0, x_3 = 0.$$

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$$\Rightarrow (x_1, x_2, x_3) = (0, 0, 0)$$

$$\Rightarrow x_1 = 0, x_2 = 0, x_3 = 0.$$

Thus $\vec{e}_1, \vec{e}_2, \vec{e}_3$ are linearly independent.

Linear Dependence and Independence of Vectors

Example

Let's show that the vectors

$\{\vec{v}_1 = (1, 2, 0), \vec{v}_2 = (2, 0, 1), \vec{v}_3 = (3, 2, 1)\} \subset \mathbb{R}^3$ are linearly dependent.

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$$\Rightarrow \begin{cases} x_1 + 2x_2 + 3x_3 = 0 \\ 2x_1 + 2x_3 = 0 \\ x_2 + x_3 = 0 \end{cases}$$

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$$\Rightarrow \begin{cases} x_1 + 2x_2 + 3x_3 = 0 \\ 2x_1 + 2x_3 = 0 \\ x_2 + x_3 = 0 \end{cases} \Rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}$$

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$$\Rightarrow x_1 + x_3 = 0, x_2 + x_3 = 0 \Rightarrow \text{If } x_3 = r, \text{ then } x_1 = x_2 = -r.$$

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$$\Rightarrow x_1 + x_3 = 0, x_2 + x_3 = 0 \Rightarrow \text{If } x_3 = r, \text{ then } x_1 = x_2 = -r.$$

This means that the system has infinitely many solution. So

$\vec{v}_1, \vec{v}_2, \vec{v}_3$ are linearly dependent.

Linear Combination of Vectors

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$$\vec{w} = x_1 \vec{v}_1 + x_2 \vec{v}_2 + \dots + x_n \vec{v}_n,$$

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then we say $\vec{w}$ is a **linear combination** of the vectors

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then we say $\vec{w}$ is a **linear combination** of the vectors

$\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$. Here, again $\vec{w} = x_1 \vec{v}_1 + x_2 \vec{v}_2 + \dots + x_n \vec{v}_n$ is an element of V .

Linear Combination of Vectors

Example

Show that $\vec{w} = (9, 2, 7) \in \mathbb{R}^3$ is a linear combination of the vectors $\vec{u} = (1, 2, -1)$ and $\vec{v} = (6, 4, 2)$, but $\vec{w}' = (4, -1, 8)$ is not.

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$$\vec{w} = x_1 \vec{u} + x_2 \vec{v}$$

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$$\vec{w} = x_1 \vec{u} + x_2 \vec{v}$$

$$(9, 2, 7) = x_1 (1, 2, -1) + x_2 (6, 4, 2)$$

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Show that $\vec{w} = (9, 2, 7) \in \mathbb{R}^3$ is a linear combination of the vectors $\vec{u} = (1, 2, -1)$ and $\vec{v} = (6, 4, 2)$, but $\vec{w}' = (4, -1, 8)$ is not.

$$\vec{w} = x_1 \vec{u} + x_2 \vec{v}$$

$$(9, 2, 7) = x_1 (1, 2, -1) + x_2 (6, 4, 2)$$

$$\Rightarrow \begin{cases} x_1 + 6x_2 = 9 \\ 2x_1 + 4x_2 = 2 \\ -x_1 + 2x_2 = 7 \end{cases}$$

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Show that $\vec{w} = (9, 2, 7) \in \mathbb{R}^3$ is a linear combination of the vectors $\vec{u} = (1, 2, -1)$ and $\vec{v} = (6, 4, 2)$, but $\vec{w}' = (4, -1, 8)$ is not.

$$\vec{w} = x_1 \vec{u} + x_2 \vec{v}$$

$$(9, 2, 7) = x_1 (1, 2, -1) + x_2 (6, 4, 2)$$

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$$\vec{w} = x_1 \vec{u} + x_2 \vec{v}$$

$$(9, 2, 7) = x_1 (1, 2, -1) + x_2 (6, 4, 2)$$

$$\Rightarrow \begin{cases} x_1 + 6x_2 = 9 \\ 2x_1 + 4x_2 = 2 \\ -x_1 + 2x_2 = 7 \end{cases} \Rightarrow \begin{pmatrix} 1 & 6 & 9 \\ 2 & 4 & 2 \\ -1 & 2 & 7 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

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Example

Show that $\vec{w} = (9, 2, 7) \in \mathbb{R}^3$ is a linear combination of the vectors $\vec{u} = (1, 2, -1)$ and $\vec{v} = (6, 4, 2)$, but $\vec{w}' = (4, -1, 8)$ is not.

$$\vec{w} = x_1 \vec{u} + x_2 \vec{v}$$

$$(9, 2, 7) = x_1 (1, 2, -1) + x_2 (6, 4, 2)$$

$$\Rightarrow \begin{cases} x_1 + 6x_2 = 9 \\ 2x_1 + 4x_2 = 2 \\ -x_1 + 2x_2 = 7 \end{cases} \Rightarrow \begin{pmatrix} 1 & 6 & 9 \\ 2 & 4 & 2 \\ -1 & 2 & 7 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow x_1 = -3, x_2 = 2.$$

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Show that $\vec{w} = (9, 2, 7) \in \mathbb{R}^3$ is a linear combination of the vectors $\vec{u} = (1, 2, -1)$ and $\vec{v} = (6, 4, 2)$, but $\vec{w}' = (4, -1, 8)$ is not.

$$\begin{aligned}\vec{w} &= x_1 \vec{u} + x_2 \vec{v} \\ (9, 2, 7) &= x_1 (1, 2, -1) + x_2 (6, 4, 2) \\ \Rightarrow \begin{cases} x_1 + 6x_2 = 9 \\ 2x_1 + 4x_2 = 2 \\ -x_1 + 2x_2 = 7 \end{cases} &\Rightarrow \begin{pmatrix} 1 & 6 & 9 \\ 2 & 4 & 2 \\ -1 & 2 & 7 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \\ \Rightarrow x_1 = -3, x_2 = 2. &\end{aligned}$$

This means that the system has a unique solution and we have

$$(9, 2, 7) = -3(1, 2, -1) + 2(6, 4, 2).$$

Linear Combination of Vectors

Example

Show that $\vec{\mathbf{w}} = (9, 2, 7) \in \mathbb{R}^3$ is a linear combination of the vectors $\vec{\mathbf{u}} = (1, 2, -1)$ and $\vec{\mathbf{v}} = (6, 4, 2)$, but $\vec{\mathbf{w}}' = (4, -1, 8)$ is not.

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$$\vec{\mathbf{w}}' = x_1 \vec{\mathbf{u}} + x_2 \vec{\mathbf{v}}$$

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Show that $\vec{\mathbf{w}} = (9, 2, 7) \in \mathbb{R}^3$ is a linear combination of the vectors $\vec{\mathbf{u}} = (1, 2, -1)$ and $\vec{\mathbf{v}} = (6, 4, 2)$, but $\vec{\mathbf{w}}' = (4, -1, 8)$ is not.

$$\begin{aligned}\vec{\mathbf{w}}' &= x_1 \vec{\mathbf{u}} + x_2 \vec{\mathbf{v}} \\ (4, -1, 8) &= x_1 (1, 2, -1) + x_2 (6, 4, 2)\end{aligned}$$

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Show that $\vec{w} = (9, 2, 7) \in \mathbb{R}^3$ is a linear combination of the vectors $\vec{u} = (1, 2, -1)$ and $\vec{v} = (6, 4, 2)$, but $\vec{w}' = (4, -1, 8)$ is not.

$$\vec{w}' = x_1 \vec{u} + x_2 \vec{v}$$

$$(4, -1, 8) = x_1 (1, 2, -1) + x_2 (6, 4, 2)$$

$$\begin{cases} x_1 + 6x_2 = 4 \\ 2x_1 + 4x_2 = -1 \\ -x_1 + 2x_2 = 8 \end{cases}$$

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Show that $\vec{w} = (9, 2, 7) \in \mathbb{R}^3$ is a linear combination of the vectors $\vec{u} = (1, 2, -1)$ and $\vec{v} = (6, 4, 2)$, but $\vec{w}' = (4, -1, 8)$ is not.

$$\begin{aligned}\vec{w}' &= x_1 \vec{u} + x_2 \vec{v} \\ (4, -1, 8) &= x_1 (1, 2, -1) + x_2 (6, 4, 2) \\ \left\{ \begin{array}{l} x_1 + 6x_2 = 4 \\ 2x_1 + 4x_2 = -1 \\ -x_1 + 2x_2 = 8 \end{array} \right. &\Rightarrow \begin{pmatrix} 1 & 6 & 4 \\ 2 & 4 & -2 \\ -1 & 2 & 8 \end{pmatrix}\end{aligned}$$

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Show that $\vec{w} = (9, 2, 7) \in \mathbb{R}^3$ is a linear combination of the vectors $\vec{u} = (1, 2, -1)$ and $\vec{v} = (6, 4, 2)$, but $\vec{w}' = (4, -1, 8)$ is not.

$$\begin{aligned}\vec{w}' &= x_1 \vec{u} + x_2 \vec{v} \\ (4, -1, 8) &= x_1 (1, 2, -1) + x_2 (6, 4, 2) \\ \left\{ \begin{array}{l} x_1 + 6x_2 = 4 \\ 2x_1 + 4x_2 = -1 \\ -x_1 + 2x_2 = 8 \end{array} \right. &\Rightarrow \begin{pmatrix} 1 & 6 & 4 \\ 2 & 4 & -2 \\ -1 & 2 & 8 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\end{aligned}$$

Linear Combination of Vectors

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This means that the system is inconsistent. Thus $\vec{w}'$ can not be written as a combination of the vectors $\vec{u}$ and $\vec{v}$.

Linear Combination of Vectors

Example

For which values of k , $\vec{w} = (1, -2, k) \in \mathbb{R}^3$ can be written as a linear combination of the vectors $\vec{u} = (3, 0, -2)$ and $\vec{v} = (2, -1, 5)$.

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$$(1, -2, k) = x_1 (3, 0, -2) + x_2 (2, -1, 5)$$

$$\begin{cases} 3x_1 + 2x_2 = 1 \\ -x_2 = -2 \\ -2x_1 + 5x_2 = k \end{cases}$$

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Linear Dependence and Independence of Vectors

Theorem

Let $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m$ be m linearly independent vectors in V . If $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m, \vec{v}_{m+1}$ are linearly dependent, then $\vec{v}_{m+1}$ can be written as a linear combination of the vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m$.

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Theorem

For $r < m$, if r vectors among $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m \in V$ are linearly dependent, then $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m$ are also linearly dependent.

Linear Dependence and Independence of Vectors

Theorem

Let V a vector space, and for $m \leq n$, $\vec{v}_1 = (a_{11}, a_{12}, \dots, a_{1n})$,
 $\vec{v}_2 = (a_{21}, a_{22}, \dots, a_{2n})$, $\dots$, $\vec{v}_m = (a_{m1}, a_{m2}, \dots, a_{mn})$ such that

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}.$$

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among m vectors are linearly independent.

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3. If $n < m$, then $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m$ are linearly dependent.

Linear Dependence and Independence of Vectors

Example

Let $\vec{\mathbf{a}} = (1, 0, 0, 1)$, $\vec{\mathbf{b}} = (0, -1, 2, 1)$, $\vec{\mathbf{c}} = (1, 2, 2, 1)$ and $\vec{\mathbf{d}} = (-2, 1, 0, 0) \in \mathbb{R}^4$ are given. Then

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1. Determine, whether or not the vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ are linearly independent or dependent?

$$\begin{array}{cccc} 1 & 0 & 0 & 1 \\ 0 & 1 & -2 & -1 \\ 1 & 2 & 2 & 1 \\ -2 & 1 & 0 & 0 \end{array}$$

$$\begin{array}{cccc} 1 & 0 & 0 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 2 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{array}$$

$$\begin{array}{cccc} 1 & 0 & 0 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 1 \end{array}$$

Linear Dependence and Independence of Vectors

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Let $\vec{\mathbf{a}} = (1, 0, 0, 1)$, $\vec{\mathbf{b}} = (0, -1, 2, 1)$, $\vec{\mathbf{c}} = (1, 2, 2, 1)$ and $\vec{\mathbf{d}} = (-2, 1, 0, 0) \in \mathbb{R}^4$ are given. Then

1. Determine, whether or not the vectors $\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$, $\vec{\mathbf{c}}$, $\vec{\mathbf{d}}$ are linearly independent or dependent?
2. Express $\vec{\mathbf{u}} = (1, -1, 2, 1)$ as a linear combination of $\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$, $\vec{\mathbf{c}}$, $\vec{\mathbf{d}}$.

Linear Dependence and Independence of Vectors

Solution (1)

$$c_1 \vec{a} + c_2 \vec{b} + c_3 \vec{c} + c_4 \vec{d} = (0, 0, 0, 0)$$

Linear Dependence and Independence of Vectors

Solution (1)

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$$\begin{aligned} \Rightarrow \quad & c_1 \quad \quad \quad + c_3 \quad - 2c_4 = 0 \\ & -c_2 \quad + 2c_3 \quad + c_4 = 0 \\ & 2c_2 \quad + 2c_3 \quad \quad \quad = 0 \\ & c_1 \quad + c_2 \quad + c_3 \quad \quad \quad = 0 \end{aligned}$$

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$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

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$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow c_1 = c_2 = c_3 = c_4 = 0$$

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is the unique solution of the system. Hence $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ are linearly independent.

Linear Dependence and Independence of Vectors

Solution (2)

$$c_1 \vec{a} + c_2 \vec{b} + c_3 \vec{c} + c_4 \vec{d} = (1, -1, 2, 1)$$

Linear Dependence and Independence of Vectors

Solution (2)

$$c_1 \vec{a} + c_2 \vec{b} + c_3 \vec{c} + c_4 \vec{d} = (1, -1, 2, 1)$$

$$\Rightarrow \begin{array}{rclcl} c_1 & & +c_3 & -2c_4 & = 1 \\ & -c_2 & +2c_3 & +c_4 & = -1 \\ & 2c_2 & +2c_3 & & = 2 \\ c_1 & +c_2 & +c_3 & & = 1 \end{array}$$

Linear Dependence and Independence of Vectors

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$$\Rightarrow [A|\mathbf{b}] = \begin{pmatrix} 1 & 0 & 1 & -2 & 1 \\ 0 & -1 & 2 & 1 & -1 \\ 0 & 2 & 2 & 0 & 2 \\ 1 & 1 & 1 & 0 & 1 \end{pmatrix}$$

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$$\Rightarrow c_1 = 0, c_2 = \frac{6}{7}, c_3 = \frac{1}{7}, c_4 = -\frac{3}{7}$$

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So we have $\frac{6}{7} \vec{b} + \frac{1}{7} \vec{c} - \frac{3}{7} \vec{d} = (1, -1, 2, 1)$.

Spanning Sets

Spanning Sets

Definition

Let $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r\} \subset V$ be given. The set spanned by S is denoted by $\text{span}(S)$ or $\langle S \rangle$ and defined as the set of possible all linear combinations of S .

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$$\text{span}(S) = \{k_1 \vec{v}_1 + k_2 \vec{v}_2 + \dots + k_r \vec{v}_r \mid k_1, k_2, \dots, k_r \in \mathbb{R}\}.$$

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Example

The spanning set of the vector $\vec{u} = (1, -2, 1) \in \mathbb{R}^3$ is

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$$\text{span}(\{(1, -2, 1)\}) = \{k(1, -2, 1) \mid k \in \mathbb{R}\}.$$

Example

The spanning set of the set

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$$\text{span}(\{(1, -2, 1, 3), (0, 2, -1, 0)\})$$

$$= \{ a(1, -2, 1, 3) + b(0, 2, -1, 0) \mid a, b \in \mathbb{R} \}.$$

?